

Understanding CFLs

Although the use of CFLs is steadily spreading, public understanding about how to dispose of them responsibly has not kept pace.

by **Richard Benware**



Candelabra CFL

The current lighting trend around the country is to incorporate compact fluorescent lamps, or CFLs, into residences and workplaces. Many energy audits have listed the use of these lightbulbs as the lowest-cost, fastest-payback way to reduce energy consumption. While CFLs cost more than conventional incandescent lightbulbs, they use about 75% less energy and last up to 10 times longer. Therefore, the market share for CFLs has recently dramatically increased to 11%, driven by advances in the technology, consumer interest, and a lower price point.

Although more and more people are using CFLs, public understanding about how to choose them, where to use them, and how to dispose of them responsibly has not kept pace. This knowledge gap is especially true when it comes to the subject of mercury. Recently, news agencies amplified a story about Brandy Bridges, of Ellsworth, Maine. A cleaning company provided her with a quote of \$2,000 to have her house cleaned after she broke a CFL. What they aren't telling you is how easy it would have been for her to clean up the bulb herself by following a few simple steps. So how can you cut through the stories to save money and energy responsibly?

Life Cycle Benefits

In order to disprove the myths about CFLs, let's begin at the beginning. When CFLs are created, manufacturers

dose the bulb with a small amount of mercury. This mercury, when electrically stimulated, releases UV light, which subsequently reacts with a phosphor coating to create visible light. Thus mercury is an essential part of every CFL; without it, the bulbs would not produce light. The typical dose of mercury is about the size of a pen tip, and these doses have been getting smaller and smaller. One reason for this is that the laws resulting from the Restriction of Hazardous Substances (RoHS) Directive have made it illegal for CFLs in Europe to contain more than 5 milligrams (mg) of mercury.

In the United States, there are no such laws limiting the amount of mercury in lightbulbs as yet, but members of the National Electrical Manufacturers Association (NEMA) have voluntarily agreed to limit the amount of mercury in the CFLs that they produce to 5 mg for bulbs of up to 25 watts and 6 mg for bulbs of 25 to 40 watts. The average CFL on store shelves today contains about 4 mg of mercury, and nearly all the CFLs in production contain less than 5 mg. The mercury used in all the CFLs produced in the United States represents 0.18% of the mercury used in all U.S. products and industrial processes.

CFLs do not release mercury as long as they are intact. In fact, they reduce net mercury emissions in the environment by conserving energy. For every kWh of electricity used by consumers, the average power plant

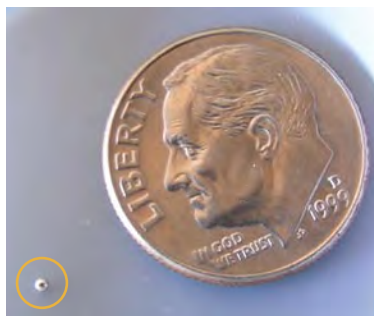
emits over 1.5 lb of pollutants. If a 75W incandescent is replaced by an 18W CFL, the CFL will use 456 kWh less energy than the incandescent over its 8,000 hour lifetime. The Emissions and Generation Resource Integrated Database (eGRID) contains data on the emissions of the average power plant. Using eGRID's information to calculate the average emissions per kWh, we find that this single CFL has prevented the release of 2.72 lb of sulfur dioxide, 1.05 lb of nitrogen oxide, 5.67 mg of mercury, and over 700 lb of CO₂.

It is important to note that these are the reductions from the average U.S. power plant. The eGRID data show that, on average, nonbaseload emissions tend to be dirtier. And in addition to reducing emissions, CFLs save money for the consumer. The Energy Information Administration (EIA) gives a 2006 average residential electricity cost of 10.08¢/kWh. Using the example given above, and basing our calculation on this figure, we find that a consumer would save about \$46 on energy over the lifetime of the CFL.

When these bulbs finally do reach the end of their useful life, there are several pathways they can take. In the best-case scenario, the bulbs are recycled. Recycling rates are increasing, thanks to state regulations—California and Minnesota have banned altogether throwing CFLs in the trash—and improved consumer awareness. In 1999, it was estimated that only 15% of all fluorescent lightbulbs were



A-Line CFL



EPA

Mercury is an essential part of every CFL; without it, the bulbs would not produce light. The typical dose of mercury is about the size of a pen tip (see circled item in photo), and these doses have been getting smaller and smaller.

recycled. Currently, that number has increased to around 25%, with higher levels in commercial applications. Since an average of 98.9% of the mercury is successfully recovered in the recycling process, this pathway generates minimal emissions.

Even the CFLs that are discarded in the trash are unlikely to release much of their mercury. Although most of them break under current trash disposal methods, some remain unbroken, and will not release any mercury. But those that do break are not likely to release much mercury. EPA estimates that only 0.2% of the remaining mercury in a spent bulb is elemental vapor. The rest of the mercury is in the glass, the phosphor coating, and the electrodes of the bulb. Mercury absorbed in these areas is not readily released. In fact, an EPA study found that only 6.8% of the total mercury in a broken bulb will be released. Since the average bulb on the market today contains only 4 mg of mercury, it will release only about 0.27

mg, even if it breaks when it is thrown in the trash.

The only disposal option that could lead to the release of any significant amount of mercury is incineration. Today, many incinerators have advanced mercury control technologies. CFLs disposed of in such incinerators would release up to 90% of their mercury, but those emissions would then be removed by these technologies. Incinerators without these technologies are not capable of removing the mercury. But even after accounting for all of the emissions that occur via all of the routes listed above, CFLs represent a mere 0.01% of total U.S. mercury emissions annually.

It is important to note that even if CFLs released all of their mercury, the environment would still be better off than it would be if nobody used CFLs. This is true because the average power plant releases 5.67 mg of mercury to power each 75W incandescent bulb. In short, replacing incandescents with CFLs is a great way to save energy, reduce mercury emissions, and save money (see "Discounting CFLs").

Choosing the Right CFL

An even better way to save energy and reduce mercury emissions is to install Energy Star-qualified light fixtures. These fixtures help consumers save energy because they use CFLs that provide at least 50, 60, or 70 lumens per watt, depending on the size of the bulb. In addition, Energy Star light fixtures use nonregressive technology. In other words, they are pin based, so

that consumers will not screw in an incandescent lightbulb when it comes time to replace the original CFL. The reason these fixtures are better at reducing mercury emissions is that pin-based CFLs must have a life exceeding 10,000 hours for the fixture to meet Energy Star specifications. Screw-based CFLs only have to have a life exceeding 6,000 hours. Since pin-based CFLs are replaced even less often than screw-based CFLs, they emit even less mercury into the environment. Plus, with pin-based lamps, the ballast is re-used, further reducing waste.

While reducing energy consumption and mercury emissions is important, contractors should know that CFLs are challenging incandescents when it comes to aesthetics, too. While the word "fluorescent" was once synonymous with "flicker" and "buzz," that was true only of the old magnetic ballasted bulbs. New, improved electronic ballasts let today's CFLs turn on instantly, and eliminate the flicker and annoying buzzing sound. In fact, to earn the Energy Star label, CFL light fixtures are required to have electronic ballasts that start in less than one second and remain illuminated.

And when it comes to color, CFLs can now produce a whole spectrum, depending on the phosphors used in the bulb. Bulbs that produce light in the 3,500°K–6,500°K range emit a higher proportion of blue wavelengths. The light produced by these bulbs is typically described as "cool," "natural light," or "daylight." Bulbs that produce light in the 2,700°K–3,000°K range emit a higher proportion of red and yellow wavelengths. The light produced by these bulbs is typically described as "warm," like the light produced by an

Discounting CFLs

Economists place a lower value on savings in the future than they place on savings in the present. This comes from the notion that money today can be put into a bank or other investment and will earn interest. The Office of Management and Budget (OMB) sets the discount rate at 7% because this rate most closely approximates the average return on private investment. At this interest rate, one could argue that \$1.07 a year from now is worth only \$1 today. Following this logic, and using OMB's discount rate, what are the discounted energy savings when 75W incandescent is replaced with an 18W CFL?

The 18W CFL used in this example has a life of 8,000 hours. Assuming average use, this would translate into a life of about 5.5 years, but I will use 6 years for the sake of simplicity. If the bulb is used equally over the 6 years, the 456 kWh energy savings would be realized at the rate of 76 kWh per year. Instead of using the Energy Information Agency (EIA's) average 2006 electricity cost of 10.08¢/kWh, I will use the more conservative value of expected future average rate, which is 9.3¢/kWh. If we multiply this rate by 76 kWh per year, we get \$7.068 in yearly energy costs avoided. Over 6 years, this discounted savings would be

$$\begin{aligned} & \$7.068 + \$7.068/1.07 + \$7.068/1.14 + \$7.068/1.22 + \$7.068/1.31 \\ & + \$7.068/1.40 \approx \$36.05 \end{aligned}$$

Thus, the actual savings would probably fall somewhere between the \$36.05 calculated here and the \$46 calculated using EIA's figures.

incandescent bulb. With all of these improvements in quality and color, it is getting harder and harder to tell the difference between light produced by CFLs and the light produced by incandescents.

Then there's the question of shape. If you think that the spiral shape of a CFL won't work for your bathroom vanity, or for your outdoor fixture, think again. Many CFLs nowadays are not spiral shaped. Several lines of covered bulbs are available. One such line is the globe-shaped bulb—a round covered CFL that looks great in locations where the bulb is visible. Another covered line is the candelabra bulb—a bulb shaped like a candle flame with a small base. These bulbs are used for chandeliers, wall fixtures, and covered outdoor fixtures.

For those who are emotionally attached to the shape of the incandescent bulb, there are CFL A-line bulbs that are nearly identical. They are used in the same applications as covered globes. There are also CFL reflector bulbs for both indoor and outdoor use. The former are designed to take the heat of recessed fixtures and track lighting, while the latter are not damaged by exposure to the elements.

As if energy savings, emissions reductions, and pleasing aesthetics weren't enough, CFLs are also easy to use. In fact, the new GU24-base fixtures and bulbs are even less of a hassle than the old screw-based fixtures and incandescent bulbs. Bulbs with the GU24 base have a twist-and-lock mechanism that makes them far quicker to install than a standard screw-based bulb. Like screw-based CFLs, these bulbs have integrated ballasts; a bulb of any wattage can be used in any fixture. Thus, consumers can change the lighting in any room with a single twist.

For a room with a dimmable fixture, there are CFLs designed specifically to replace dimmable incandescent bulbs. The newest of these bulbs have a mechanism that automatically shuts off the bulb when a threshold of about 10% light level is reached. This extends the life of the bulb under consistent dimming conditions and prevents it from flickering before going out.

Basically, if the fixture uses an incandescent, it can use a CFL. With energy-efficient bulbs that can do so many things, there is no reason to use their incandescent alternative.



Spiral CFL

What the Laws Say

Several states currently have legislation under consideration regarding the restriction of incandescent bulbs. California took the lead by proposing a bill to require an energy efficiency standard on the sale of general-purpose lightbulbs by 2012. Since very few, if any, incandescent bulbs can currently meet this standard, or will likely meet the standard by 2012, it will effectively constitute a ban. Connecticut, Rhode Island, and North Carolina have followed suit by proposing similar legislation. Other states have taken a step in this direction by considering legislation that requires CFLs to be used in government buildings by 2010. Bills of this kind are being considered by New Jersey, South Carolina, Illinois, and Hawaii.

Taking a different approach, Minnesota is considering a bill to place a \$0.25 tax on the sale or transfer of incandescent bulbs. Two entire countries—Canada and Australia—have recently announced

their intention of banning inefficient incandescents. With so many governments seeking to ban or decrease the use of incandescent bulbs, it may be in contractors' and distributors' best interests to ensure that they have a supply chain to procure CFLs.

In addition to knowing these laws, it is important to know which states require the recycling of CFLs. In California and Minnesota, it's illegal to throw CFLs into the trash. It is also illegal in Ottawa, though Canada as a whole still has only voluntary recycling programs. In this country, EPA encourages recycling any compact fluorescent, regardless of state laws, on the reasoning that doing so reduces the mercury load on the environment and also eliminates some waste. To find the nearest recycling center that accepts CFLs, visit the Web sites listed under "For more information."

CFLs are a dominant technology. They are more efficient than incandescent bulbs, they last much longer than incandescent bulbs, and they function just as well as incandescent bulbs, if not better. It is in the world's best interest for everyone—from contractors to homeowners—to use them. The average home has 45 lightbulbs in 30 fixtures. Simply switching to CFLs can reduce the average home's waste and emissions. Recycling these bulbs at the end of their useful life can reduce these emissions further. All it takes is a quick trip to the store to find all of the great products available for your needs.

H
e

Richard Benware is a graduate student at Cornell University, in Ithaca, New York. He recently researched CFLs for Energy Star in EPA's Student Temporary Employment program (STEP).

lighting

For more information:

For information on Energy Star products, including CFLs, go to www.energystar.gov.

To determine which type of CFL would best replace your incandescent, visit www.energystar.gov/cfls.

For more on how to dispose of fluorescent lightbulbs, go to www.lamprecycle.org or www.energystar.gov/CFLsandMercury.

To find a fluorescent lightbulb recycling center near you, visit www.earth911.org or www.epa.gov/bulbrecycling.

For directions on how to clean up a broken CFL, go to www.epa.gov/CFLcleanup.

National Electrical Manufacturers Association
1300 N 17th St., Ste. 1752
Rosslyn, VA 22209
Tel: (703)841-3200
Web site: www.nema.org.